The first law of thermodynamics

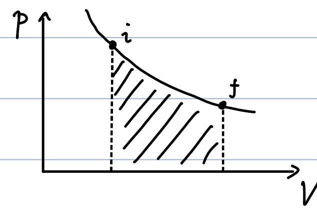
Quasi-static process :

Slow enough $\longrightarrow$ system remains in thermal equilibrium at all times

Work

$$\text{đ}W = F ds = pA ds \boxed{= p dV} \qquad (\text{recall last class})$$

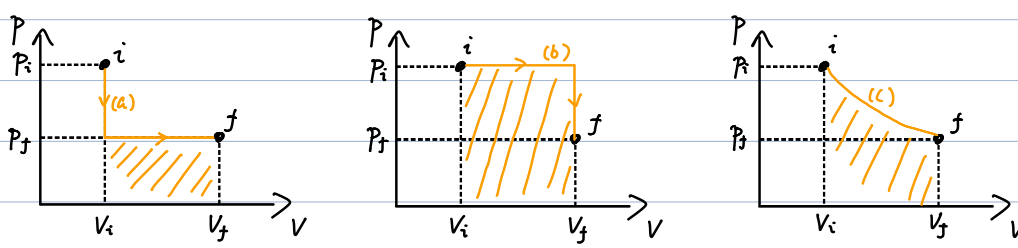
$$\Rightarrow \quad W = \int p dV$$



Sign of W (depends on definition) :

- Work done BY the gas : $\text{đ}W = p dV$
- Work done TO the gas : $\text{đ}W = -p dV$

W depends on path :



$$W^{(a)} = P_f (V_f - V_i) \qquad W^{(b)} = P_i (V_f - V_i) \qquad W^{(a)} < W^{(c)} < W^{(b)}$$

The first law of thermodynamics

$$\Delta U = \Delta Q - W$$

ΔQ: Heat absorbed W: work done by the system

Differential form :

$$dU = \text{đ}Q - \text{đ}W$$

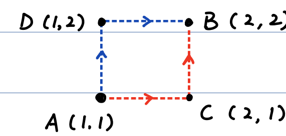
đ : path dependent

Inexact / Imperfect differential :

corresponding integral is path-dependent.

Example.

$$\bar{d}g = dx + \frac{x}{y}dy \qquad (\text{inexact})$$



Path-1 : $A \to C \to B$

$$\left(\int_A^C + \int_C^B \right) \left(dx + \frac{x}{y} dy \right)$$

$$= 1 + 2\ln 2$$

Path-2 : $A \to D \to B$

$$\left(\int_A^D + \int_D^B \right) \left(dx + \frac{x}{y} dy \right)$$

$$= \ln 2 + 1$$

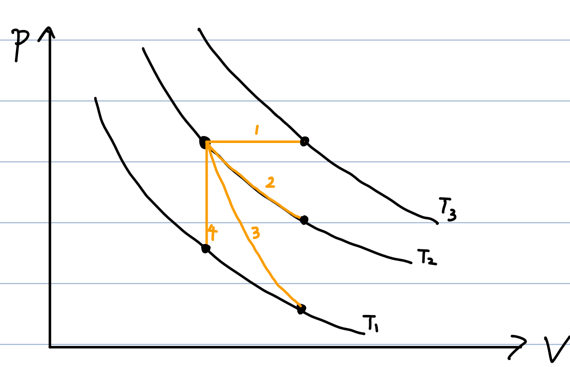
$$df \equiv \frac{\bar{d}g}{dx} = \frac{dx}{x} + \frac{dy}{y} \qquad (\text{exact})$$

For both paths :

$$\int_A^B df = 2\ln 2$$

Different processes (NOT necessarily quasi-static)

- Isobaric : p fixed
- Isochoric (Isovolumetric) : V fixed
- Isothermal : T fixed
- Adiabatic : $Q = 0$ (no heat transfer)
- ...



1 : isobaric

2 : Isothermal

3 : Adiabatic

4 : Isochoric

Quasi-static isothermal expansion :

Work done by gas $W = \int_{V_i}^{V_f} p \, dV$

$$= \int_{V_i}^{V_f} \frac{nRT}{V} dV$$

$$= nRT \ln \frac{V_f}{V_i}$$

Quasi-static adiabatic process :

$\boxed{pV^\gamma = \text{constant}}$ $\boxed{\gamma \equiv C_p / C_v}$

Proof :

$$\begin{cases} dU = \text{đ}Q - p\,dV \\ \qquad\ \ 0 \text{ (adiabatic)} \\ U = nc_v T \end{cases} \qquad c_v = \frac{f}{2} R$$

$\Rightarrow \quad nc_v\, dT = -p\,dV \qquad ①$

$pV = nRT$

$\Rightarrow \quad p\,dV + V\,dp = nR\,dT \qquad ②$

$① ② \Rightarrow \quad -\frac{p}{c_v} dV = \frac{p}{R} dV + \frac{V}{R} dp$

$\Rightarrow \quad p\left(\frac{1}{R} + \frac{1}{c_v}\right) dV + \frac{V}{R} dp = 0$

$\Rightarrow \quad \left(1 + \frac{R}{c_v}\right) \frac{dV}{V} + \frac{dp}{p} = 0$

$\Rightarrow \quad \frac{c_p}{c_v} \frac{dV}{V} + \frac{dp}{p} = 0 \qquad c_p = c_v + R$

$\Rightarrow \quad \gamma \ln V + \ln p = \text{constant}$

$\Rightarrow \quad pV^\gamma = \text{constant} \qquad$ proved.

Adiabatic free expansion

- No heat $\Delta Q = 0$ (adiabatic)
- No work $W = 0$ (free)
- Not quasi-static (out of equilibrium except for initial & final states)
- $U_i = U_f \Rightarrow T_i = T_f$

Heat engine (abbr "engine")

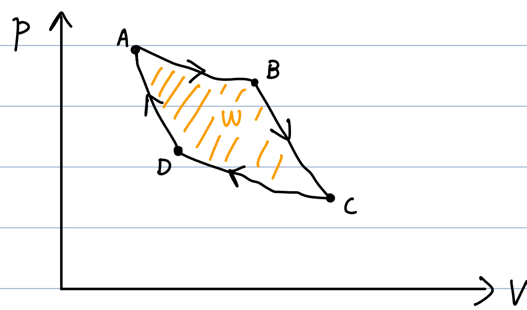
Extracts energy (heat) from environment and does work.

e.g. Steam engine

Gasoline engine

...

cyclic process in engine



$A \to B \to C \to D \to A$

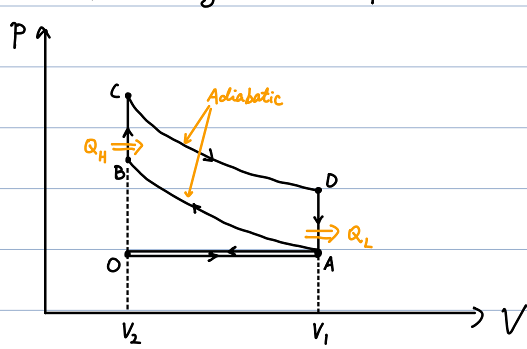
$$\begin{cases} \Delta U = 0 \\ W = \oint p\,dV \end{cases}$$

$$\Rightarrow \underbrace{\Delta Q}_{\text{Net heat absorbed}} = W = \oint p\,dV$$

Example. Otto cycle

1876, Nicolaus Otto

(commonly used in petrol engines)



$O \rightleftharpoons A$ often omitted in schematic drawings.

Q_H: Heat absorbed from High-T reservoir

Q_L: Heat released to Low-T reservoir

Four Strokes:

$O \to A$: Intake

$A \to B$: Compression

$B \to C \to D$: Combustion (Ignition) (Power)

$D \to A \to O$: Exhaust

Thermal efficiency:

$$\boxed{\varepsilon \equiv \frac{W}{Q_H}}$$

$$W = Q_H - Q_L$$

$$\Rightarrow \boxed{\varepsilon = 1 - \frac{Q_L}{Q_H}}$$

For Otto cycle:

$$\begin{cases} Q_L = n c_v (T_D - T_A) \\ Q_H = n c_v (T_C - T_B) \end{cases}$$

$$\Rightarrow \quad \varepsilon_{Otto} = 1 - \frac{T_D - T_A}{T_C - T_B}$$

Furthermore, for adiabatic process:

$$PV^\gamma = \text{constant} \qquad\qquad PV = nRT$$

$$\Rightarrow \quad TV^{\gamma-1} = \text{constant}$$

$A \to B$: $T_A V_1^{\gamma-1} = T_B V_2^{\gamma-1}$

$$\Rightarrow \left(\frac{V_2}{V_1}\right)^{\gamma-1} = \frac{T_A}{T_B}$$

$C \to D$: $T_C V_2^{\gamma-1} = T_D V_1^{\gamma-1}$

$$\Rightarrow \left(\frac{V_2}{V_1}\right)^{\gamma-1} = \frac{T_D}{T_C}$$

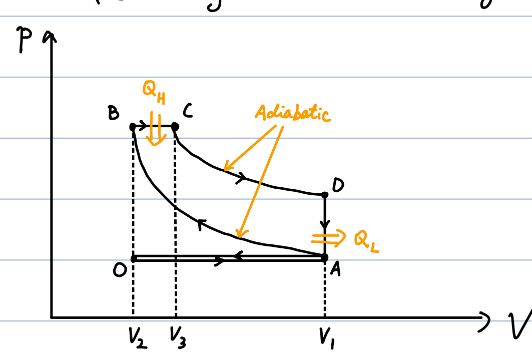
$\Rightarrow$

$$\varepsilon_{Otto} = 1 - \frac{T_D - T_A}{T_C - T_B}$$

$$= 1 - \left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

Example. Diesel cycle

(commonly used in diesel engines)



$$T_A V_1^{\gamma-1} = T_B V_2^{\gamma-1}$$

$$\Rightarrow \boxed{T_B = \left(\frac{V_1}{V_2}\right)^{\gamma-1} T_A}$$

$$\begin{cases} P_B V_2 = nRT_B \\ P_C V_3 = nRT_C \\ P_B = P_C \end{cases}$$

$$\Rightarrow \boxed{T_C} = \frac{V_3}{V_2} T_B = \boxed{\frac{V_3}{V_2}\left(\frac{V_1}{V_2}\right)^{\gamma-1} T_A}$$

$$T_D V_1^{\gamma-1} = T_C V_3^{\gamma-1}$$

$$\Rightarrow \boxed{T_D} = \left(\frac{V_3}{V_1}\right)^{\gamma-1} T_C = \left(\frac{V_3}{\cancel{V_1}}\right)^{\gamma-1} \frac{V_3}{V_2}\left(\frac{\cancel{V_1}}{V_2}\right)^{\gamma-1} T_A = \boxed{\left(\frac{V_3}{V_2}\right)^{\gamma} T_A}$$

$$nc_v(T_C - T_B) = Q_H - P_B(V_3 - V_2)$$

$$\Rightarrow \boxed{Q_H} = nc_v(T_C - T_B) + P_B(V_3 - V_2)$$

$$= nc_v(T_C - T_B) + nR(T_C - T_B)$$

$$= \boxed{nc_p(T_C - T_B)}$$

$$\boxed{Q_L = nc_v(T_D - T_A)}$$

$\Rightarrow$ Efficiency of diesel cycle :

$$\boxed{\varepsilon_{diesel}} = 1 - \frac{Q_L}{Q_H}$$

$$= 1 - \frac{c_v(T_D - T_A)}{c_p(T_C - T_B)}$$

$$= 1 - \gamma^{-1}\frac{(V_3/V_2)^{\gamma} - 1}{(V_3/V_2)(V_1/V_2)^{\gamma-1} - (V_1/V_2)^{\gamma-1}}$$

$$= \boxed{1 - \gamma^{-1}\frac{(V_3/V_2)^{\gamma} - 1}{(V_3/V_2) - 1}\left(\frac{V_2}{V_1}\right)^{\gamma-1}}$$

Example. Dual cycle

(commonly used in ship engines)